WORKED SOLUTION

STEP I 2007, Question 14

Probability and Statistics • Poisson Parameter Uniqueness

STEP I 2007, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s1-q14

The key idea

Each piece of information gives an equation of the form $h(\lambda) = \text{constant}$, where h is some function of the mean. Whether that equation pins down λ uniquely depends entirely on the *shape* of h on $\lambda > 0$: a strictly monotonic h hits each value at most once, whereas an h that rises and then falls hits most values twice and only its maximum value once. So the whole question is about finding where each function is monotonic and where it peaks.

Solution

Throughout, $\lambda > 0$ (a Poisson mean is positive) and $\mathrm{P}(X = k) = \mathrm{e}^{-\lambda}\lambda^k/k!$ for $k = 0, 1, 2, \ldots$

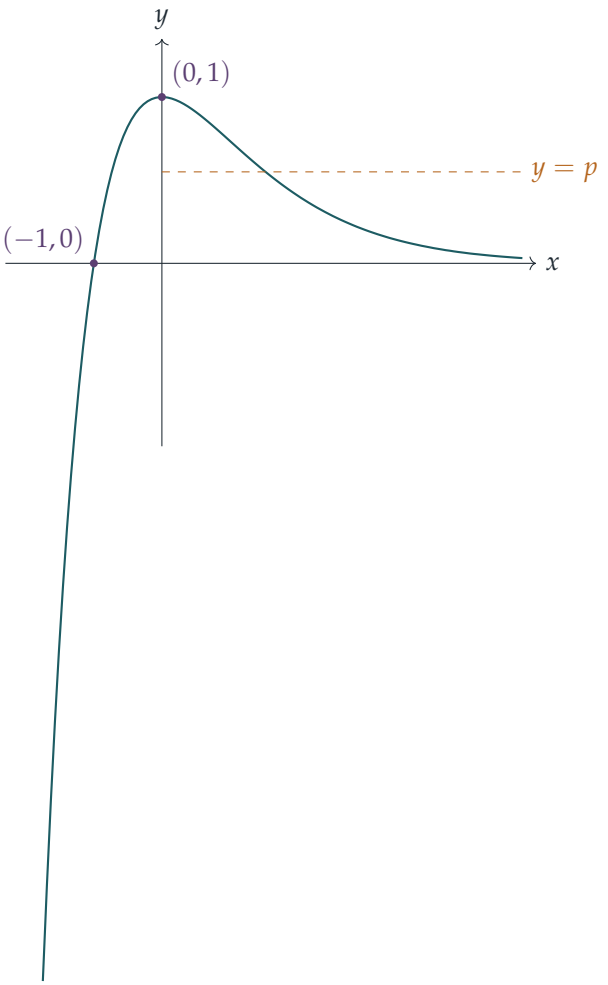
Part (i)

Let $f(x) = (x+1)\mathrm{e}^{-x}$. By the product rule,

$$f'(x) = \mathrm{e}^{-x} - (x+1)\mathrm{e}^{-x} = -x\,\mathrm{e}^{-x}.$$

Since $\mathrm{e}^{-x} > 0$, $f'(x)$ has the opposite sign to x: f is increasing for $x < 0$ and decreasing for $x > 0$. So the only turning point is a maximum at $(0, 1)$, which is also where the curve crosses the y-axis. The curve meets the x-axis where $x + 1 = 0$, at $(-1, 0)$.

For the ends: as $x \to \infty$, e^{-x} beats the linear factor and $f(x) \to 0$ from above (the x-axis is an asymptote); as $x \to -\infty$, both factors grow in size and $f(x) \to -\infty$.



Now the probability. We have

$$\mathrm{P}(X \geq 2) = 1 - \mathrm{P}(X = 0) - \mathrm{P}(X = 1) = 1 - \mathrm{e}^{-\lambda} - \lambda\mathrm{e}^{-\lambda} = 1 - f(\lambda),$$

so the information says exactly that

$$f(\lambda) = (1 + \lambda)\mathrm{e}^{-\lambda} = p.$$

On $\lambda > 0$ the function f is continuous and strictly decreasing (since $f'(\lambda) = -\lambda\mathrm{e}^{-\lambda} < 0$), falling from $f(0) = 1$ towards 0 as $\lambda \to \infty$. So it takes every value in $(0, 1)$ exactly once: by the intermediate value theorem there is at least one solution, and strict monotonicity rules out a second. In the sketch, the horizontal line $y = p$ with $0 < p < 1$ meets the right-hand branch exactly once. Hence λ is uniquely determined.

Part (ii)

Here $\mathrm{P}(X = 1) = \lambda\mathrm{e}^{-\lambda} = q$. Write $g(\lambda) = \lambda\mathrm{e}^{-\lambda}$. Then

$$g'(\lambda) = (1 - \lambda)\mathrm{e}^{-\lambda},$$

so g is strictly increasing on $(0,1]$ and strictly decreasing on $[1,\infty)$, with maximum value $g(1)=\mathrm{e}^{-1}$. Also $g(\lambda)\to 0$ as $\lambda\to 0^+$ and as $\lambda\to\infty$. (In fact $g(\lambda)=\mathrm{e}^{-1}f(\lambda-1)$, so the graph of g is the curve from part (i) shifted one unit right and scaled vertically by e^{-1}.)

Ask yourself How many times can a horizontal line meet a curve that rises to a peak and then falls?

Each monotonic piece is hit at most once, so:

- if $0<q<\mathrm{e}^{-1}$, there is one solution in $(0,1)$ and another in $(1,\infty)$, so two possible values of λ;
- if $q>\mathrm{e}^{-1}$, there are no solutions;
- if $q=\mathrm{e}^{-1}$, the only solution is the peak itself, $\lambda=1$.

So the information determines a unique λ for exactly one value of q, namely

$$q=\mathrm{e}^{-1},\qquad \lambda=1.$$

Part (iii)

By the definition of conditional probability, and since the event $\{X=1\}$ lies inside $\{X\le 2\}$,

$$\mathrm{P}(X=1\mid X\le 2)=\frac{\mathrm{P}(X=1)}{\mathrm{P}(X\le 2)}=\frac{\lambda\mathrm{e}^{-\lambda}}{\mathrm{e}^{-\lambda}\left(1+\lambda+\frac{1}{2}\lambda^2\right)}=\frac{2\lambda}{2+2\lambda+\lambda^2}.$$

Call this $k(\lambda)$. Dividing top and bottom by λ (allowed, as $\lambda>0$),

$$k(\lambda)=\frac{2}{\lambda+\dfrac{2}{\lambda}+2}.$$

Ask yourself What is the smallest value of $\lambda+2/\lambda$ for $\lambda>0$?

By the AM–GM inequality, $\lambda+\dfrac{2}{\lambda}\ge 2\sqrt{\lambda\cdot\dfrac{2}{\lambda}}=2\sqrt{2}$, with equality exactly when $\lambda=2/\lambda$, that is $\lambda=\sqrt{2}$. Moreover $\lambda+2/\lambda$ is strictly decreasing on $(0,\sqrt{2}]$ and strictly increasing on $[\sqrt{2},\infty)$ (its derivative $1-2/\lambda^2$ changes sign only at $\sqrt{2}$), and it tends to ∞ at both ends. So k has the same rise-then-fall shape as g in part (ii), with maximum

$$k(\sqrt{2})=\frac{2}{2\sqrt{2}+2}=\frac{1}{\sqrt{2}+1}=\sqrt{2}-1.$$

Exactly as before: for $0<r<\sqrt{2}-1$ there are two values of λ, for $r>\sqrt{2}-1$ there are none, and for $r=\sqrt{2}-1$ there is exactly one. So λ is uniquely determined only when

$$r=\sqrt{2}-1,\qquad \lambda=\sqrt{2}.$$

Alternatively, clear denominators in $k(\lambda) = r$ to get the quadratic $r\lambda^2 - 2(1-r)\lambda + 2r = 0$. Its roots have product $2 > 0$ and sum $2(1-r)/r > 0$, so any real roots are both positive; there is exactly one root precisely when the discriminant $4\left[(1-r)^2 - 2r^2\right] = 4(1 - 2r - r^2)$ vanishes, giving $r = \sqrt{2} - 1$ (taking the positive root) and the double root $\lambda = (1-r)/r = \sqrt{2}$.

Checks

- Part (ii): $\lambda = 1$ gives $\mathrm{P}(X = 1) = \mathrm{e}^{-1} \approx 0.3679$; a numerical scan of $\lambda\mathrm{e}^{-\lambda}$ over $0 < \lambda < 40$ never exceeds this, and $q = 0.3$ gives two crossings.
- Part (iii): with $\lambda = \sqrt{2}$, $\dfrac{2\sqrt{2}}{2 + 2\sqrt{2} + 2} = \dfrac{\sqrt{2}}{2 + \sqrt{2}} = \sqrt{2} - 1 \approx 0.4142$, and a scan confirms this is the maximum; $r = 0.35$ gives two values of λ, $r = 0.45$ none.
- Sanity: as $\lambda \to 0$ almost all of the mass is at 0, and as $\lambda \to \infty$ the mass given $X \leq 2$ piles up at 2, so in both limits $\mathrm{P}(X = 1 \mid X \leq 2) \to 0$, as the formula says.

Common slips

- Forgetting that $\lambda > 0$: in part (i) the curve also exists for $x < 0$, but only the branch $x > 0$ is relevant to the probability.
- Claiming uniqueness in part (i) without saying why: you need both that a solution exists (continuity, values running from 1 down to 0) and that there is no second one (strict monotonicity).
- In part (iii), using $\mathrm{P}(X = 1 \text{ and } X \leq 2) = \mathrm{P}(X = 1)\,\mathrm{P}(X \leq 2)$; the events are not independent, but $\{X = 1\}$ is contained in $\{X \leq 2\}$.

Beyond STEP

This is a first taste of *identifiability*: in statistics we ask whether an observed quantity determines the parameter of a model, and the answer is yes exactly when the map from parameter to observable is one-to-one. Here $\mathrm{P}(X \geq 2)$ identifies λ, but $\mathrm{P}(X = 1)$ and $\mathrm{P}(X = 1 \mid X \leq 2)$ generally do not. The same rise-and-fall shape reappears when you maximise a likelihood: $\lambda\mathrm{e}^{-\lambda}$ is the likelihood of λ after seeing a single observation $X = 1$, and its peak at $\lambda = 1$ is the maximum likelihood estimate. Blitzstein and Hwang, *Introduction to Probability*, is a lively place to read more about the Poisson distribution and conditioning.

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